

# Advanced Machine Learning

## Vector Spaces

$$Mx=b$$

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 & 16 \\ 1 & 3 & 9 & 27 & 81 \\ 1 & 4 & 16 & 64 & 256 \\ 1 & 5 & 25 & 125 & 625 \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 2 \\ -3 \\ 1 \end{pmatrix}$$

$$b=M^{-1}x$$

$$Mv_i = \lambda_i v_i$$

$$\text{Tr}(X^{-1}A) = -X^{-1}AX^{-1}$$

Vectors, metric spaces, norms

# Outline

1. **Vector Spaces**
2. Metrics (distances)
3. Norms

$$Mx=b$$

$$Mv_i = \lambda_i v_i$$

$$b=M \cdot$$

# Matrices, Vectors and All That

- The language of machine learning is mathematics
- Sometimes we draw pretty pictures to explain the mathematics
- Much of the mathematics we will use involves vectors, matrices and functions
- You need to master the language of mathematics, otherwise you won't understand the algorithms
- I'm going to spend this lecture and the next revising the mathematics you need to know

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# Vectors

- We often work with objects with many components (features)
- To help handle this we will use vector notation
  - We represent vectors by bold symbols
  - All our vectors are column vectors by default
  - We treat them as  $n \times 1$  matrix
- We write row vectors as transposes of column vectors

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

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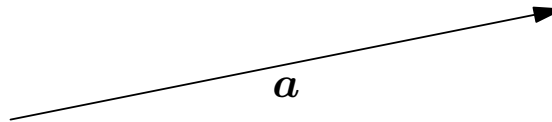
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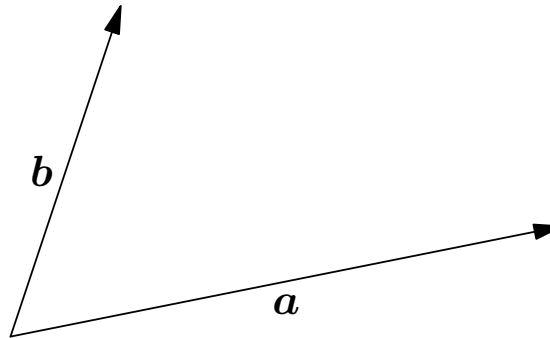
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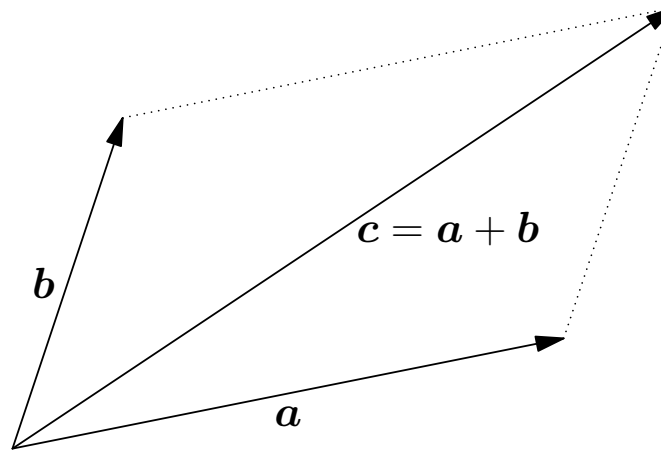
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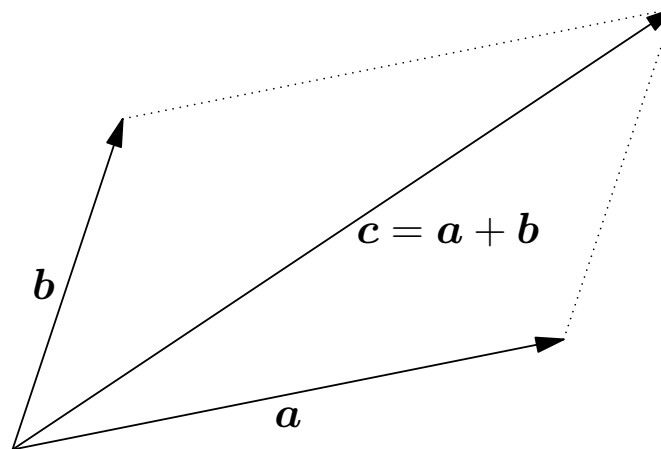
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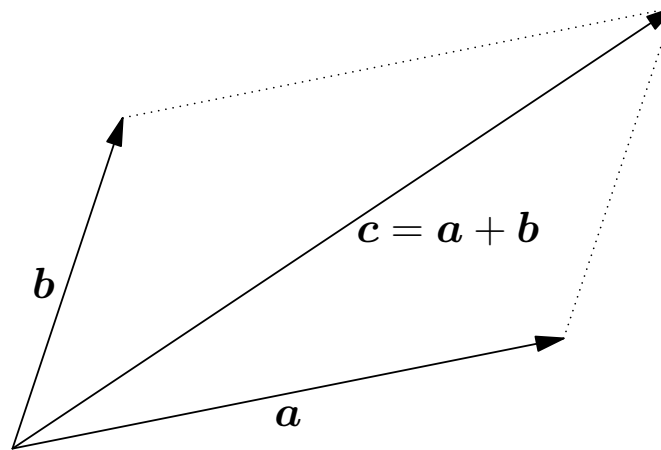
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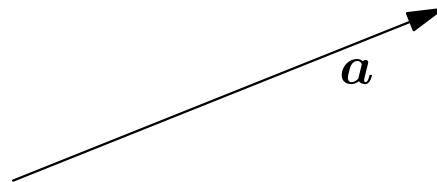
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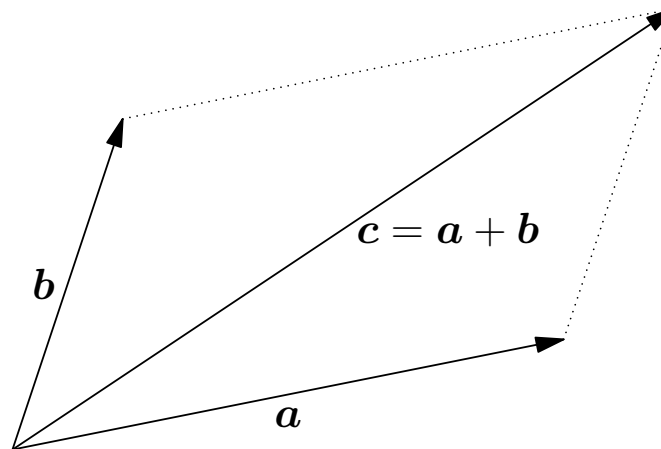


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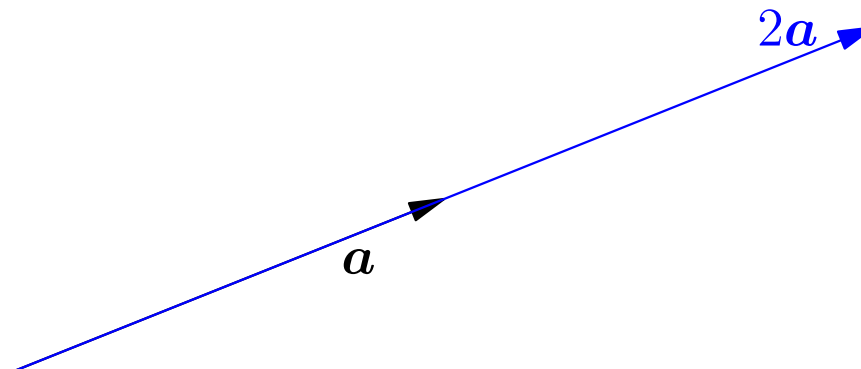


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# Vector Space

- A vector space,  $\mathcal{V}$ , is a set of vectors which satisfies

1. if  $\mathbf{v}, \mathbf{w} \in \mathcal{V}$  then  $a\mathbf{v} \in \mathcal{V}$  and  $\mathbf{v} + \mathbf{w} \in \mathcal{V}$  (closure)
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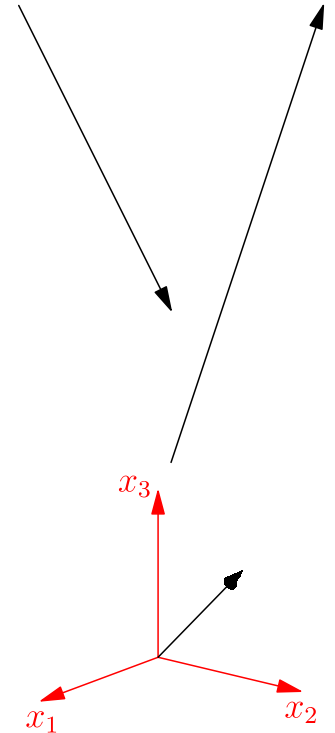
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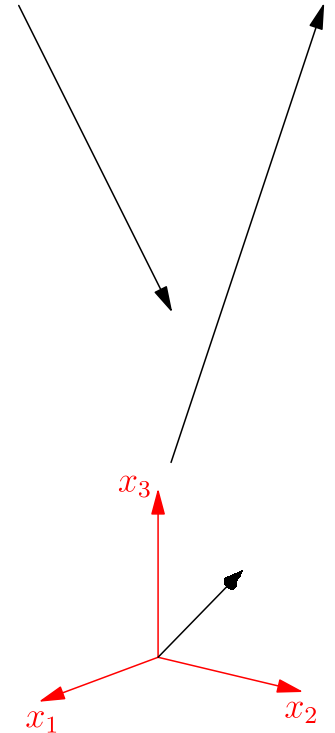
- When we first learn about vectors we think of them as arrows in 3-D space
- If we centre them all at the origin then there is a one-to-one correspondence between vectors and points in space
- We call this vector space  $\mathbb{R}^3$
- Any set of quantities  $\mathbf{x} = (x_1, x_2, \dots, x_n)^\top$  which satisfy the axioms above form a vector space  $\mathbb{R}^n$
- Of course, we can't so easily draw pictures of high-dimensional vectors





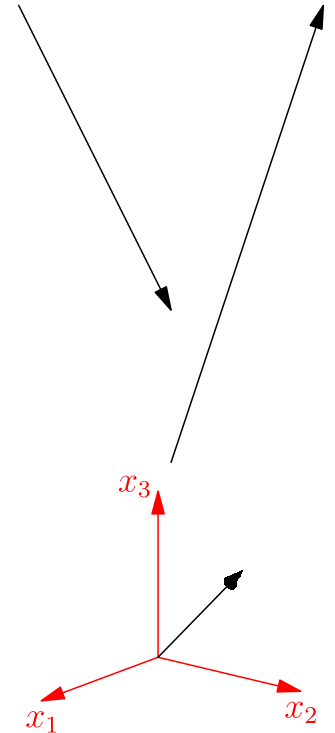
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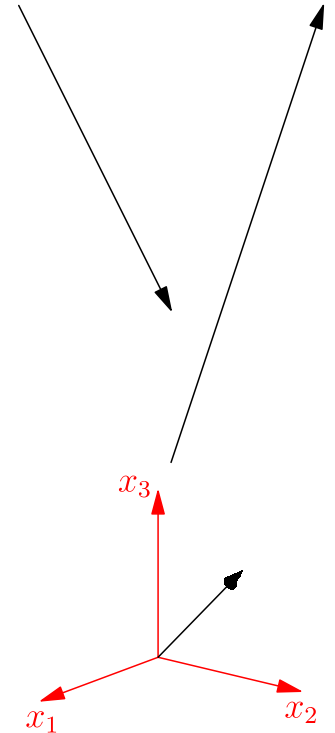
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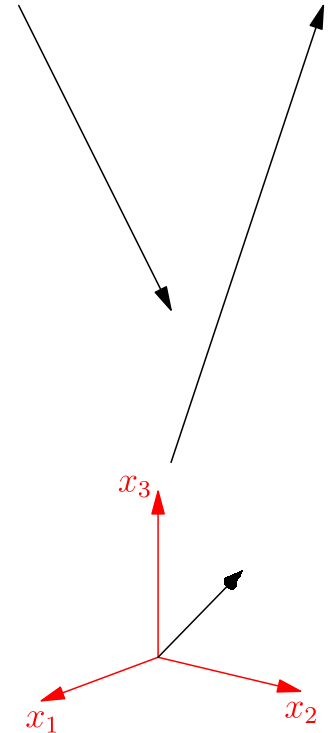
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# Other Vector Spaces

- Any set of object that satisfies the axioms of a vector spacer are *vectors*
- Matrices satisfy all the conditions of a vector space
- Infinite sequences form a vector space
- Functions form a vector space
  - ★ Let  $C(a,b)$  be the set of functions defined on the interval  $[a,b]$
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# Other Vector Spaces

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# Outline

1. Vector Spaces
2. Metrics (distances)
3. Norms

$$Mx=b$$

$$Mv_i = \lambda_i v_i$$

$$b=Mv$$

# Metrics

- Vector spaces become more interesting if we have a notion of distance
- We say  $d(\mathbf{x}, \mathbf{y})$  is a **proper distance** or **metric** if
  1.  $d(\mathbf{x}, \mathbf{y}) \geq 0$  (non-negativity)
  2.  $d(\mathbf{x}, \mathbf{y}) = 0$  iff  $\mathbf{x} = \mathbf{y}$  (identity of indiscernibles)
  3.  $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x})$  (symmetry)
  4.  $d(\mathbf{x}, \mathbf{y}) \leq d(\mathbf{x}, \mathbf{z}) + d(\mathbf{z}, \mathbf{y})$  (triangular inequality)
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- We are often interested in functions that behave nicely
- E.g. They are continuous



# Lipschitz Function

- One way to characterise well behaved function,  $f(x)$  is if there exists a number  $K < \infty$  such that for all  $x$  and  $y$

$$d(f(x), f(y)) \leq K d(x, y)$$

- This is known as a **Lipschitz condition** and the function is said to be  $K$ -Lipschitz
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# Contractive Mappings

- An interesting class of function are those for which  $K < 1$
- These are said to be contractive mappings
- A famous theorem that applies to contractive mappings is the Banach fixed-point theorem which says there exists a unique fixed point such that  $f(x) = x$
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# Outline

1. Vector Spaces
2. Metrics (distances)
3. **Norms**

$$Mx=b$$

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$$b=M$$

# Norms

- Vector spaces are even more interesting with a notion of length
- **Norms** provide some measure of the size of a vector
- To formalise this we define the **norm** of an object  $v$  as  $\|v\|$  satisfying
  1.  $\|v\| > 0$  if  $v \neq \mathbf{0}$  (non-negativity)
  2.  $\|av\| = a\|v\|$  (linearity)
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# Vector Norms

- The familiar vector norm is the (Euclidean) two norm

$$\|\mathbf{v}\|_2 = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$$

- Other norms exist, such as the  $p$ -norm ( $p \geq 1$ )

$$\|\mathbf{v}\|_p = \left( \sum_{i=1}^n |v_i|^p \right)^{1/p}$$

- Special cases include the 1-norm and the infinite norm

$$\|\mathbf{v}\|_1 = \sum_{i=1}^n |v_i| \qquad \|\mathbf{v}\|_\infty = \max_i |v_i|$$

- The 0-norm is a pseudo-norm as it does not satisfy condition 2

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# Matrix Norms

- We can define norms for other objects
- The norm of a matrix encodes how large the mapping is
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# Compatible Norms

- A vector and matrix norm are said to be compatible if

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(Spectral and Euclidean norms are compatible)

- Norms provide quick ways to bound the maximum growth of a vector under a mapping induced by the matrix
- We will see that a measure of the sensitivity of a mapping is in terms of the ratio of its maximum effect to its minimum effect on a vector
- This is known as the **conditioning**, given by  $\|\mathbf{M}\| \times \|\mathbf{M}^{-1}\|$

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# Why Should You Care?

- Deep learning involves multiply the input (which we can think of as a vector  $\mathbf{x}$ ) by many layers
- In CNNs we have convolutional layers and dense layers
- The effect of applying these layers can be represented by a matrix multiplication  $\mathbf{x}_n = \mathbf{L}_n \mathbf{x}_{n-1}$
- We also do other things like applying ReLU's or pooling that changes the magnitude,  $\mathbf{x}_n$ , of our representation
- If you are developing new architectures you want  $\|\mathbf{x}_n\|$  neither to blow up or vanish
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- We also do other things like applying ReLU's or pooling that changes the magnitude,  $\mathbf{x}_n$ , of our representation
- If you are developing new architectures you want  $\|\mathbf{x}_n\|$  neither to blow up or vanish
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# Function Norms

- Functions can also have norms, for example, if  $f(x)$  is defined in some interval  $\mathcal{I}$

$$\|f\|_{L_2} = \sqrt{\int_{x \in \mathcal{I}} f^2(x) dx}$$

- The  $L_2$  vector space is the set of function where  $\|f\|_{L_2} < \infty$
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# Summary

- Vector spaces with a distance (metric spaces) and vector spaces with a norm (normed vector spaces) are interesting objects
- They allow you to define a topology (open/closed sets, etc.)
- You can build up ideas about connectedness, continuity, contractive maps, fixed-point theorems, . . .
- For the most part we are going to consider an even more powerful vector space that has an inner-product defined

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